GDNet: Naming Security Products from the First Five Packets of a Flow

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Abstract

We present GDNet, a compact byte-level classifier that names the security product behind a network flow from its first five packets [1], [7]. Unlike signature-based fingerprinting [17], [19] and learned application classifiers [8], [9], it targets the security-agent inventory: the antivirus, endpoint-detection, VPN, web-application-firewall, and monitoring products installed on a host. A fixed 1,600-byte vector is formed from the first five packets with the IPv4 addresses masked [1], and four Gated DeltaNet blocks with a shared 256-dimensional CLS readout feed four task heads and a prototype gate [2], [4], [11]. On a 503-flow test split the four-arm ensemble reaches 0.8787 product accuracy, 0.9245 tier accuracy, and 0.9784 gate AUROC while rejecting all 18 unknown-product bins. An exact byte-level audit of six public benchmarks shows that five of them share approximately 80% of their test flows with their training splits, whereas ISCXTor2016 shares none [14], [15], so deduplicated numbers are reported alongside the original ones and no state-of-the-art claim is made.

Index Terms: byte-level traffic classification, security product identification, asset discovery, tokenizer-free learning, Gated DeltaNet, open-set recognition, dataset leakage, encrypted traffic analysis

I Introduction

Endpoint agents are installed by operations, VPN endpoints and the web application firewall by the network team, and the NDR sensor by whoever owns monitoring, so no single owner holds the consolidated product view for a host [17], [19], [21]. Asset records are populated at provisioning time and reconciled rarely afterwards, so a host that has been reimaged, reassigned, or fitted with a new agent can carry a product set that differs from the recorded entry [14], [16].

Port and protocol rules do not close that gap. The label set used here is closed at 18 products and services, listed in Appendix A with their type and tier, and it contains three VPN products (wireguard, openvpn, and stunnel) that accept operator-configured ports [19], three web-facing products (squid_proxy, ntopng, and guacamole_mgmt) that share 80 and 443 in the default configuration [19], and four endpoint agents (windows_defender, clamav_freshclam, velociraptor, and osquery) whose update traffic follows the same HTTPS pattern [20], [21]. Rule-based fingerprinting resolves the operating system from header quirks [17], the protocol from payload patterns [19], or the TLS stack from a handshake hash [18], whereas learned device identification resolves the device type from flow statistics [20], [21]. Neither produces a product label from a closed set of security agents, and neither supplies an abstain option calibrated on held-out sessions [11], [12].

GDNet addresses that setting with a compact byte-level model. The input is the first five packets of a flow, packed into 1,600 bytes with the IPv4 addresses zeroed [1], [7]. The encoder is four Gated DeltaNet blocks of $d_{model} = 256$ and $d_{state} = 16$, giving 2.86M parameters per arm [2], [3], [4]. The classifier is a four-arm ensemble of 11.44M parameters that occupies 46 MB on disk [1], [8], [9]. A separate prototype gate decides whether the product head may be trusted [11], [12], [13], so the system returns a product label, a type, a tier, a prototype distance, and a confidence, or an explicit unknown.

For a flow f, with P the closed set of 18 products listed in Appendix A, T the set of product types, and G the set of tiers, the classifier returns a tuple $(\hat{y}, \hat{t}, \hat{g}, d, s)$ with $\hat{y}$ in P or unknown, $\hat{t}$ in T, $\hat{g}$ in G, a prototype distance d, and a confidence s [11], [13]. Rejection is a threshold of $\tau = 0.1041$, the 68th percentile of development distances, applied to the distance between the flow embedding and the nearest of three class prototypes [12], [13]. It is not an additional output class, so no unknown-class examples are required during training, which matters because the set of products absent from a given network cannot be enumerated in advance [5], [6].

On the 40,283 labelled flows used here the ensemble reaches 0.8787 product accuracy on a 503-flow test half, with 0.9245 tier accuracy and 0.9784 gate AUROC, and it rejects all 18 unknown-product bins in the acceptance set [14], [16]. The probe is a single C file of 42 KB and peak VRAM is 3.40 GB at batch 128 on one RTX 4090 [28], [29], so the inference path fits on the card that also runs the analyst's tooling. The same property makes the result sensitive, because an adversary who can enumerate the defensive stack of a host can select techniques that avoid it, and Section VI-E states the release constraints that follow [14], [22].

This paper makes four contributions.

- We introduce GDNet, a byte-level classifier that assigns flows to 18 security products and an explicit unknown class from the first five packets, using 2.86M parameters per arm and 11.44M for the four-arm ensemble, with no tokenizer and no externally pre-trained corpus [1], [6], [9].
- We describe a rejection mechanism based on three prototypes per class and a threshold calibrated on held-out sessions, which rejects all 18 unknown-product bins in the acceptance set without training on unknown samples [11], [12].
- We report an exact byte-level audit of six public traffic benchmarks, five of which share approximately 80% of their test flows with their training splits, and we quantify the effect of deduplication on the three affected benchmarks that were in scope [14], [15], [16].
- We release an inference stack with checksums that reproduces the reported numbers without redistributing any capture [28], [29].

II Related Works

II-A Traffic fingerprinting and asset discovery

p0f infers an operating system from TCP/IP header characteristics such as the initial TTL and the ordering of TCP options, using a hand-written signature table and no training data [17]; nDPI classifies protocols by matching payload and header patterns at line rate [19]; and JA3 with JA3S hash the fields of the TLS ClientHello and ServerHello into short strings that identify client and server implementations [18]. Learned device identification moves one step further: IoT Sentinel trains a random forest over header features to name device types across a small device vocabulary [21], and Sivanathan et al. separate smart-home and campus device classes on flow statistics alone [20]. All of these stop at the operating system, the protocol, or the device type, and none produces a product label from the closed set used here or supplies an abstain option.

II-B Byte-level traffic classification

ET-BERT tokenizes datagrams before pre-training a Transformer with masked language modeling [7], YaTC renders flows as images and pre-trains a masked autoencoder over patches [8], and NetMamba applies a unidirectional Mamba backbone to packet sequences [9]; all three require a pre-training stage over an external corpus, and all three descend from the byte-level and subword conventions established for text [5], [6]. MambaNetBurst removes both the tokenizer and the pre-training stage and learns directly from raw bytes with a compact Mamba-2 stack of 2.7M parameters [1], which is the closest architectural relative of the present work. GDNet keeps the byte vocabulary, the CLS readout, and the fixed-length burst convention of that design and replaces the mixer with four Gated DeltaNet blocks [4], which retain the linear cost of the state-space formulation [2], [3] while adding a delta-rule update to the transition.

II-C Foundation models and open-set recognition

netFound learns packet representations from a large token corpus and reports frozen-encoder results alongside explicit corrections for dataset shortcuts, at roughly two orders of magnitude more parameters than one 2.86M-parameter GDNet arm [10]. Open-set recognition treats rejection as a decision separate from classification, since softmax confidence is a poor boundary estimator [13] and distance to class prototypes is a calibrated alternative [11], [12]. We keep three sub-prototypes per class inside 54 prototype rows per gate arm, define distance as one minus the maximum cosine similarity, and calibrate τ on the development half only.

II-D Dataset leakage

Trustee showed that network machine-learning systems exploit dataset shortcuts including packet-level splits and timestamp correlations [14], and Pcap-Encoder demonstrated that per-packet splits leak TCP sequence numbers so that a model can score highly without learning the task [15]. netFound reports corrected splits for several public datasets [10], and NetBench proposes a common evaluation protocol [16]. Section V-D extends this line with exact byte-level duplicate counts over 6 benchmarks and 250,072 flows, computed without reference to any model.

TABLE I. Placement of GDNet among fingerprinting, learned classification, and foundation-model approaches.

Method	Representation	Tokenizer	Pre-training	Sequence model	Target
p0f [17]	TCP/IP header fields	none	none	signature table	OS
nDPI [19]	payload + header	none	none	DPI rules	protocol
JA3/JA3S [18]	TLS hello fields	none	none	hash	client/server
IoT Sentinel [21]	header features	none	none	random forest	device type
ET-BERT [7]	datagram token	BERT vocab	masked LM	Transformer	application
YaTC [8]	flow image patch	patch	masked AE	Transformer	application
NetMamba [9]	packet token	vocab	self-supervised	Mamba	application
MambaNetBurst [1]	raw byte	none	none	Mamba-2	application
netFound [10]	packet token	vocab	large corpus	Transformer	multiple
GDNet	raw byte	none	contrastive, in-domain	GDN	security product

III Architecture

III-A Input representation

Each flow is reduced to a fixed vector of 1,600 bytes: five packets of up to 320 bytes, of which the first 80 are headers and the next 240 are payload, with bytes 12 through 19 holding the IPv4 addresses zeroed before the vector leaves the probe while ports, protocol numbers, flags, and payload content are preserved [1], [7].

$$x = [\, b_1, b_2, \ldots, b_{1600} \,],\ b_i \in \{0, \ldots, 255\} \tag{1}$$

Shorter flows are zero-padded and longer flows truncated at the same 1,600-byte boundary, Ethernet, VLAN, and raw IPv4 framing are all accepted, and parsing never sits on the hot path because the probe emits the vector as soon as the fifth packet of a flow arrives [28]. The five-packet budget carries the transport handshake and the first application message, which is the segment in which the 18 products of Appendix A disclose a version string, a certificate, or a fixed update header, whereas an address is a property of a site rather than of a product [17], [19], [21].

III-B Byte embedding, backbone, and heads

Each byte is mapped to a 256-dimensional embedding, a learnable CLS token is appended, positional information is added, and four GDN blocks process the sequence, whose final CLS state feeds four heads while a training-only domain head loads with strict=False and is discarded at inference [1], [4]. The four heads together hold about 0.4M of the 2.86M parameters, so the encoder accounts for 86% of the model, and the malicious head is trained on public malware traffic and reported separately because that corpus does not match the collection conditions of any particular site.

TABLE II. Backbone configuration of one arm.

Component	Setting
Byte embedding	256
GDN blocks	4, d_{model} = 256, d_{state} = 16
CLS token	learnable, appended after the last byte
Feature	final CLS state, 256 dims
State dict entries	103, of which 52 are GDN block entries
Parameters per arm	2.86M

TABLE III. Task heads, about 0.4M parameters in total.

Head	Layers	Outputs	In main results
Product	256-512-18	18	yes
Tier	256-3 linear	3	yes
Type	256-512-10	10	yes
Malicious	256-512-11	11	reported separately

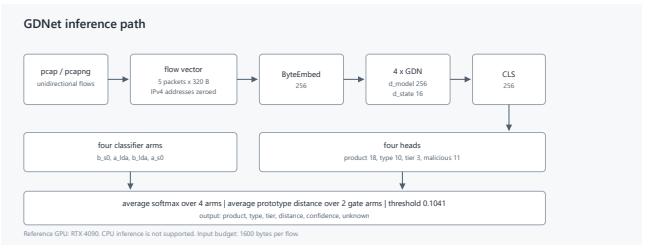


Fig. 1. Inference path of one arm.

III-C Open-set decision

Three prototypes per class are produced by KMeans over the training embeddings, giving 54 prototype rows per gate arm, and for a feature f and the prototype set P_c of class c the distance and the decision rule are

$$d(f) = 1 - \max_{c,k} \cos(f, p_{c,k}),\ p_{c,k} \in P_c \tag{2}$$

$$\hat{y} = \operatorname{argmax}_c s_c(f) \text{ if } \bar{d}(f) \le \tau, \text{ otherwise unknown} \tag{3}$$

where s is the averaged softmax vector, $\bar{d}$ the distance averaged over the two gate arms, and τ = 0.1041 the 68th percentile of development distances [11], [12], [13]. Because the cosine distance is bounded by the range [0, 2], the threshold is a quantile of the development distribution rather than an absolute similarity, and any new site requires its own recalibration since the threshold is a property of the traffic rather than of the model.

III-D Ensemble, complexity, and footprint

Six arms were trained, formed by crossing two pre-training runs A and B with three training variants: a standard multi-task objective (S0), logit adjustment by class prior (LDA), and hard-negative weighting inside the contrastive term (HN). Four were retained, A-S0, A-LDA, B-LDA and B-S0, while A-HN and B-HN reached a higher AUROC yet their gate distance moved whenever the threshold was adjusted, so they were dropped on stability grounds rather than on accuracy [1], [9]; averaging is unweighted.

TABLE IV. The six trained arms. Released checkpoints carry the same labels in lower case, for example stage3_a_s0.pth for A-S0.

Arm	Run	Training variant	Role in release
A-S0	A	standard multi-task	classifier
A-LDA	A	logit adjustment	classifier and gate
B-LDA	B	logit adjustment	classifier and gate
B-S0	B	standard multi-task	classifier
A-HN	A	hard-negative weighting	dropped
B-HN	B	hard-negative weighting	dropped

$$s(f) = (1/4)\,\Sigma_{j=1..4}\, softmax(\, z_j(f)\,),\ \bar{d}(f) = (1/2)\,\Sigma_{j\,\in\,\{A\text{-}LDA,\ B\text{-}LDA\}} \quad (4)$$
$$d_j(f)$$

A Gated DeltaNet block is linear in sequence length, so a forward pass costs $O(L \cdot d_{model} \cdot d_{state})$ with L = 1,600 [2], [3], [4], and memory is dominated by activations rather than weights, with 46 MB of parameters against 3.40 GB of peak VRAM at batch 128.

```
Algorithm 1: Inference for a single flow
Input: packet stream, threshold τ
       arms S = {A-S0, A-LDA, B-LDA, B-S0}
Output: product, type, tier, distance, confidence, or
unknown

 1  collect the first five packets of a unidirectional flow
 2  x <- 1600-byte vector with bytes 12:20 zeroed
 3  for each arm j in S:
 4      e_j <- GDN_encoder_j(x)                 # 256-d CLS
 5      s_j <- softmax(product_head_j(e_j))     # 18 logits
 6  s <- mean over j of s_j
 7  d <- mean over gate arms of 1 - max cosine(e, prototype)
 8  if d <= τ: return argmax s, type head, tier head, d, max
s
 9  else:      return unknown, d
```

IV Experiments

IV-A Data, splits, and implementation

The labelled set contains 40,283 flows over 18 classes captured on laboratory hosts and virtual machines, each product having been exercised through its normal update, telemetry, or management path rather than through synthetic payloads [7], [8], [30], [32]. The split is at flow level rather than session level, and because each class appears in exactly one held-out session the test figures describe those sessions rather than a new organisation; 14 sessions cover 18 classes because a single capture session exercises several products on one host [14], [16].

TABLE V. Data, splits, and training configuration.

Item	Value
Classes	18 products and services
Labelled flows	40,283
Capture sources	laboratory hosts and virtual machines
Validation flows	1,006, from 14 held-out sessions
Dev / test split	503 / 503 at flow level
Selection	threshold, checkpoint, and ensemble on dev only
Test usage	once
Hardware	4 × RTX 4090, fp32
Stage 1	contrastive, NT-Xent, 33,184 training-only flows
Stage 2	supervised fine-tuning, 60 epochs, multi-task
Stage 3	REINFORCE, 300 epochs
Reference stack	Python 3.12, PyTorch 2.14

The training code is not in the release because it assumes the laboratory data layout of 40,283 flows and would require cleanup before another group could execute it, whereas the released checkpoints, 54 prototype rows, threshold of 0.1041, and evaluation script reproduce the test-half numbers directly [28], [29]. Model selection is confined to the 503-flow development half, so the 503-flow test half is scored once [14], [16].

Training follows three stages: the contrastive stage optimises NT-Xent over 33,184 training-only flows [23], fine-tuning adds a domain-adversarial term [24] under AdamW [26] with defaults inherited from Adam [27], and the last stage applies REINFORCE to three rewards, namely correct product, correct type, and honest rejection [25].

Reproducibility has two halves: the released checkpoints, 54 prototype rows, threshold and evaluation script reproduce the test-half figures whenever the split is available, while the split itself is withheld because it derives from laboratory traffic; the probe-to-service path collected 99,888 flows in 30 seconds on the test host and the service labelled container-management traffic as unknown, which is the expected behaviour outside the product domain.

IV-B Metrics

Product accuracy counts exact product labels and tier accuracy counts exact tier labels, macro-F1 averages over the classes that have at least one test flow according to

$$F1_{macro} = (1/|C^{+}|)\,\Sigma_{c\,\in\,C^{+}}\, 2\,P_c\,R_c\,/\,(P_c + R_c) \quad (5)$$

where C^{+} denotes that set, gate AUROC compares known flows against unknown and novel flows [12], [13], and unknown rejection reports the share of unknown-product bins correctly refused, while malicious macro-F1 is computed on the deduplicated USTC-TFC2016 test split of 1,242 flows [15], [32] and accuracy intervals are Wilson intervals at 95%.

V Results

TABLE VI. Closed-set accuracy, gate quality, and rejection.

Metric	Test (503 flows)	Wilson ±	All validation (1006 flows)	Wilson ±
Product accuracy	0.8787	0.029	0.8817	0.020
Tier accuracy	0.9245	0.023	0.9334	0.016
Macro-F1, supported classes	0.7621	—	0.8240	—
Gate AUROC	0.9784	—	0.9820 (dev)	—
Unknown rejection	18/18	—	18/18	—
Malicious macro-F1, dedup USTC	—	—	0.9702	—
Malicious binary F1	—	—	>0.994	—

V-A Security product identification

The product head attains 0.8787 on the test half and 0.8817 across the full validation set, tier accuracy 0.9245 and 0.9334, and gate AUROC 0.9784 and 0.9820 [1], [9]. All 18 unknown-product bins are rejected at the fixed threshold of 0.1041, with the rejection rate unchanged between q = 0.64 and q = 0.75 [11], [12]; wireguard has a single test flow, so its class score moves between 1.0 and 0.0 on one error, and the test-half macro-F1 of 0.7621 is therefore not a headline figure [14], [16]. The all-validation value of 0.8240 is the stable reference, and the malicious head reaches 0.9702 macro-F1 on 1,242 deduplicated USTC flows without falling below 0.994 binary F1 [15], [32].

V-B Acceptance and enrollment

TABLE VII. Acceptance and enrollment on 264 product bins and 18 unknown bins.

Item	Result
Product bins	264
Unknown bins	18, all rejected
Products correct, raw ensemble	8/10
Products correct, with timing rule	9/10, rule not released
stunnel recall	0/30 at every threshold
wireguard	2 bins, treated as noise
HTTP service, FLW1 records	6/6
Enrollment, smb	8/8 after three sub-prototypes

A timing rule over inter-arrival times raises the acceptance figure from 8 to 9 of 10 and is excluded from the release because it is a second model without an independent calibration, while stunnel produces 0 hits from 30 bins at every threshold examined; enrollment on smb reaches 8 of 8 after three sub-prototypes were registered, which shows the mechanism functions but does not supply the data stunnel lacks [11], [21]. The HTTP service returns 6 of 6 known flows on FLW1 records, which is the only end-to-end check that traverses the code path a deployment would use [28], [29].

V-C Throughput and memory

TABLE VIII. Throughput and memory on one RTX 4090 in fp32, measured at batch 128 for VRAM and at batch 1 for latency.

Configuration	Throughput	Batch-1 latency	Peak VRAM, batch 128
Single arm	1,017 flows/s	10.7 ms	3.37 GB
Four-arm ensemble	253 flows/s	43.1 ms	3.40 GB
Distilled student	1,011 flows/s	10.6 ms	3.37 GB

The probe captures faster than the 253-flow/s ensemble classifies, so a deployment must sample, shard across several devices, or classify on demand, and an 8 GB card retains headroom at batch 128 with 3.40 GB of peak VRAM [28], [29]. The distilled student reaches 1,011 flows/s at 0.8231 validation accuracy, 5.6 points below the ensemble, and is kept experimental for that reason [23].

V-D Public split audit

TABLE IX. Exact byte-level train/test overlap in six public benchmarks.

Benchmark	Train	Test	Train ∩ test	Share of test	Clean test
ISCXVPN2016	13,281	1,384	1,106	79.9%	278
ISCXTor2016	11,655	1,458	0	0%	1,458
USTC-TFC2016	37,323	6,662	5,605	84.1%	1,057
CICIoT2022	18,734	1,951	1,571	80.5%	380
CrossPlatform-Android	44,700	4,656	3,733	80.2%	923
CrossPlatform-iOS	40,379	4,205	3,396	80.8%	809

The audit hashes every flow twice, first over the PNG file bytes and then over the decoded grayscale pixels, and the two hashes agree on all 250,072 flows, which rules out PNG encoding or metadata as the source of the overlap; a duplicate is an exact byte match and the unit is 1,600 bytes, the same vector the model consumes [14], [15]. Duplicate shares reach 79.9% on ISCXVPN2016, 84.1% on USTC-TFC2016, 80.5% on CICIoT2022, 80.2% on CrossPlatform-Android, and 80.8% on CrossPlatform-iOS, leaving clean test sets of 278, 1,057, 380, 923, and 809 flows respectively, whereas ISCXTor2016 duplicates none of its 1,458 test flows [30], [31], [32].

Label conflicts across splits are zero and every duplicate remains inside its class, which is consistent with independent sampling from a single flow pool without cross-split deduplication [14], [16]. The finding is not universal, because the Lens official splits carry between 7.0% and 17.2% burst overlap on the four in-scope tasks and between 0.05% and 0.5% on the two mobile tasks, with at most 57 conflicting rows against test [10], [16].

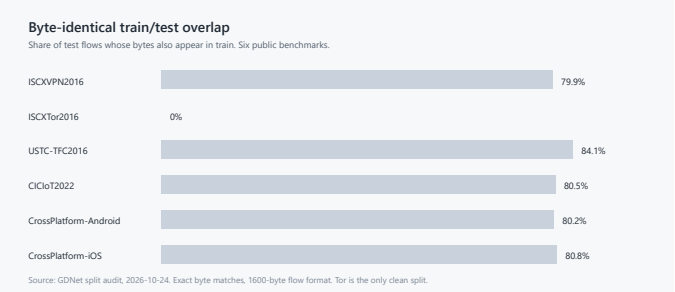


Fig. 2. Exact byte-level train/test overlap in the six audited benchmarks.

V-E Deduplicated evaluation

TABLE X. Re-evaluation on the deduplicated subsets of three affected benchmarks.

Item	VPN	USTC	CIoT
Clean test flows	278	1,057	380
Accuracy on clean set	0.9376	0.9981	0.9868
Macro-F1 on clean set	0.9418	0.9671	0.9823
Full-set macro-F1	0.9861	0.9949	0.9965
Drop	−4.4 pp	−2.8 pp	−1.4 pp

Re-scoring on the clean subsets costs 4.4 percentage points of macro-F1 on VPN, 2.8 on USTC, and 1.4 on CIoT, so the leak inflated the score without accounting for all of it, and both figures are reported here for the three benchmarks that were rerun [15], [16]. ISCXTor2016 required no rerun, and the internal comparison against the external baseline on the same clean subsets is excluded because the checkpoints it depends on are not ours to redistribute [10].

V-F Ablations and variants that did not ship

TABLE XI. Variants and ablations that were examined.

Variant	Result	Decision
Auxiliary pre-training (field, order, reconstruction)	Probe +9.35 points; downstream E1 below E0+LDA	not used
Timing rule	Acceptance 8 to 9; unstable at nearby thresholds	not released
Distillation	0.8231 validation accuracy; 5.6 points below ensemble	experimental
stunnel threshold tuning	0/30 at every threshold	needs data or enrollment
Two hard-negative arms	Higher AUROC; gate distance moved with τ	dropped

Auxiliary pre-training added field, order, and reconstruction objectives, raised the probe score by 9.35 points, and then lost to the plain E0-plus-LDA pipeline downstream, so it was dropped [6], [23]. Depth, width, state size, embedding dimension, and packet budget were fixed once from the deployment constraint and the MambaNetBurst reference rather than swept, so the available compute went to the 250,072-flow audit and the 264-bin acceptance set instead [1], [3].

VI Discussion

VI-A Five packets are enough because the handshake is where the product signs itself.

The acceptance set reaches 8 of 10 products from the first five packets alone, because each product in Appendix A emits a version string, a certificate, a fixed header, or an update request inside that budget [17], [19], [21].

VI-B Model size follows the deployment constraint.

Each arm holds 2.86M parameters and the ensemble 11.44M, which fits in 46 MB and reaches 3.40 GB of peak VRAM at batch 128, while the 42 KB probe runs in a container without a Python runtime [28], [29]; a larger model would plausibly score higher [1], [10], but the operating constraint here was the 8 GB card that also runs the analyst's tooling.

VI-C Rejection matters more than accuracy.

The gate rejects 18 of 18 unknown bins at $\tau = 0.1041$ with the operating point stable from $q = 0.64$ to $q = 0.75$ [11], [12], and without that rejection the 0.8787 product accuracy would send the responder toward software that is not installed whenever a flow falls outside the 18 classes [13], [22].

VI-D Threats to validity.

The split is flow-level with one session per class, so the absolute figures describe those 14 sessions [14], [16]; the 503-flow test set yields a Wilson interval of ±0.029 on product accuracy, and no interval is available for macro-F1 without the per-class counts; only one capture environment was used, so cross-site behaviour is unmeasured; the audit detects exact duplicates only, and covers neither near-duplicates, nor cross-dataset overlap, nor label errors inside a split [14], [15]; the label set is closed at 18 products, so a product that updates silently will be rejected or misclassified; and no seed sweep was performed.

VI-E Use and misuse.

The release contains 19 weight files totalling 94.6 MB, no captures, and SHA-256 checksums, and it names defence products, so it is run only where authorisation exists and captures stay off public demonstrations [22]. The outputs are scores rather than verdicts, the

malicious head is reported separately and excluded from automated response paths, and the input carries no addresses and no payload semantics beyond byte patterns.

VII Conclusion

GDNet names security products from the first five packets of a flow using 11.44M parameters and no tokenizer, and it declines to answer when a flow does not resemble anything in its training set, attaining on a 503-flow test split 0.8787 product accuracy, 0.9245 tier accuracy, and 0.9784 gate AUROC while rejecting all 18 unknown-product bins in the acceptance set [1], [9]. The audit is the component we would retain if only one could be kept, because five of six public traffic benchmarks share approximately 80% of their 250,072 audited flows across the train and test boundary and every figure in this paper is reported with that constraint in view [14], [15], [16].

Acknowledgments

The Gated DeltaNet kernel is taken from flash-linear-attention [29], and the public benchmarks were constructed by the groups that published them [30], [31], [32], the audit in Section V-D existing because those datasets were released in a form that permitted independent verification.

References

[1] G K Kulatilleke, S Layeghy, M Baktashmotlagh, and M Portmann, "MambaNetBurst: direct byte-level network traffic classification without tokenization or pretraining," arXiv:260511034, 2026; [2] A Gu and T Dao, "Mamba: linear-time sequence modeling with selective state spaces," arXiv:231200752, 2023; [3] T Dao and A Gu, "Transformers are SSMs: generalized models and efficient algorithms through structured state space duality," in Proc ICML, 2024; [4] S Yang, J Kautz, and A Hatamizadeh, "Gated delta networks: improving Mamba2 with the delta rule," in Proc ICLR, 2025; [5] A Vaswani et al, "Attention is all you need," in Proc NeurIPS, 2017; [6] L Xue et al, "ByT5: towards a token-free future with pre-trained byte-to-byte models," TACL, vol 10, 2022; [7] X Lin, G Xiong, G Gou, Z Li, J Shi, and J Yu, "ET-BERT: a contextualized datagram representation with pre-training transformers for encrypted traffic classification," in Proc WWW, 2022; [8] R Zhao, M Zhan, X Deng, Y Wang, Y Wang, G Gui, and Z Xue, "Yet another traffic classifier (YaTC): a masked autoencoder for flow images," arXiv:230519077, 2023; [9] NetMamba: efficient network traffic classification via pre-training unidirectional Mamba, arXiv:240511886, 2024; [10] netFound: foundation model for network security, arXiv:231017025, 2023, revised 2026; [11] J Snell, K Swersky, and R Zemel, "Prototypical networks for few-shot learning," in Proc NeurIPS, 2017; [12] W J Scheirer, A de Rezende Rocha, A Sapkota, and T E Boult, "Toward open set recognition," IEEE TPAMI, vol 35, no 7, 2013; [13] A Bendale and T E Boult, "Towards open set deep networks," in Proc IEEE CVPR, 2016; [14] N Jacobs et al, "AI/ML for network security: the emperor has no clothes," in Proc ACM CCS, 2022; [15] R Zhao et al, "The sweet danger of sugar: the pitfalls of pcap-based encoders," in Proc ACM SIGCOMM, 2025, arXiv:250716438; [16] NetBench: a benchmark for network traffic classification, in Proc ACM FMSys, 2024; [17] M Zalewski, "p0f v3," https://lcamtufcoredumpcx/p0f3/; [18] J Althouse, "TLS fingerprinting with JA3 and JA3S," Salesforce Engineering, 2017; [19] L Deri, M Martinelli, T Bujlow, and A Cardigliano, "nDPI: open-source high-speed deep packet inspection," in Proc IEEE IWCMC, 2014; [20] A Sivanathan, H H Gharakheili, F Loi, A Radford, C Wijenayake, A Vishwanath, and V Sivaraman, "Characterizing and classifying IoT traffic in smart cities and campuses," in Proc IEEE INFOCOM Workshops, 2017; [21] M Miettinen, S Marchal, I Hafeez, N Asokan, A-R Sadeghi, and S Tarkoma, "IoT Sentinel: automated device-type identification for security enforcement in IoT," in Proc IEEE ICDCS, 2017; [22] R Sommer and V Paxson, "Outside the closed world: on using machine learning for network intrusion detection," in Proc IEEE S&P, 2010; [23] T Chen, S Kornblith, M Norouzi, and G Hinton, "A simple framework for contrastive learning of visual representations," in Proc ICML, 2020; [24] Y Ganin et al, "Domain-adversarial training of neural networks," JMLR, vol 17, no 59, 2016; [25] R J Williams, "Simple statistical gradient-following algorithms for connectionist reinforcement learning," Machine Learning, vol 8, 1992; [26] I Loshchilov and F Hutter, "Decoupled weight decay regularization," in Proc ICLR, 2019; [27] D P Kingma and J Ba, "Adam: a method for stochastic optimization," in Proc ICLR, 2015; [28] A Paszke et al, "PyTorch: an imperative style, high-performance deep learning library," in Proc NeurIPS, 2019; [29] flash-linear-attention, https://githubcom/fla-org/flash-linear-attention; [30] G Draper-Gil, A H Lashkari, M S I Mamun, and A A Ghorbani, "Characterization of encrypted and VPN traffic using time-related features," in Proc ICISSP, 2016; [31] A H Lashkari, G Draper-Gil, M S I Mamun, and A A Ghorbani, "Characterization of Tor traffic using time-based features," in Proc ICISSP, 2017; [32] W Wang, M Zhu, J Wang, X Zeng, and Z Yang, "End-to-end encrypted traffic classification with one-dimensional convolution neural networks," in Proc IEEE ISI, 2017.

Appendix A: Label Set

TABLE XII. The 18 products and services used as the closed label set, with type and tier.

Product	Type	Tier
windows_defender	Antivirus	Security software
clamav_freshclam_win	Antivirus	Security software
clamav_freshclam	Antivirus	Security software
velociraptor	EDR / HIDS	Security software
velociraptor_win	EDR / HIDS	Security software
osquery	EDR / HIDS	Security software
wazuh_agent	EDR / HIDS	Security software
wireguard	Zero trust / VPN	Security appliance
openvpn	Zero trust / VPN	Security appliance
stunnel	Zero trust / VPN	Security appliance
squid_proxy	Internet behavior management	Security appliance
ntopng	NDR	Security appliance
guacamole_mgmt	Bastion host	Security appliance
waf_modsecurity	WAF	Security appliance
suricata_update	NDR	Security appliance
web_http	Browser	Productivity software
ssh	Remote administration	Productivity software
ftp	File transfer	Productivity software

No capture is redistributed in the release: the weights were trained on laboratory traffic and on the training splits of public benchmarks [30], [31], [32], and the SHA-256 checksums in the model repository allow any reader to verify which files are loaded [29].